Hardware Components Assignment

Day 3: In-Depth RAM and Storage Analysis

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Introduction to System Memory and Storage

In modern computing, both RAM (Random Access Memory) and Storage are crucial for system performance, but they serve fundamentally different purposes. RAM acts as the system's short-term, volatile memory where active data is stored for immediate access by the CPU. In contrast, storage drives (HDDs, SSDs, NVMe) provide long-term, non-volatile data retention, ensuring the operating system, applications, and user files are preserved even when the power is turned off. This assignment explores the generational advancements in these technologies and documents practical hardware manipulation.

Task 1: RAM Generation Comparison (DDR3 vs DDR4 vs DDR5)

The evolution of Double Data Rate (DDR) memory has focused on two primary goals: increasing data transfer speeds (bandwidth) and reducing power consumption (voltage). Below is a comprehensive comparison of the three most prevalent RAM generations in the current market landscape:

RAM Type	Speed Range (MT/s)	Standard Voltage	Typical Use Cases & Architecture
DDR3	800 - 2133 MT/s	1.5V (or 1.35V for low-voltage DDR3L variants)	Legacy systems, older office PCs (pre-2015), retro gaming setups, and budget NAS builds. It features a single 64-bit channel per stick.
DDR4	2133 - 3600+ MT/s	1.2V	Mainstream modern PCs, standard gaming builds, general productivity laptops, and current-generation servers. Offers higher density and faster clock speeds.
DDR5	4800 - 7200+ MT/s	1.1V	High-end contemporary gaming (4K), 3D rendering workstations, heavy multitasking, and AI development. Features dual 32-bit channels per stick and on-die ECC for better stability at high speeds.

Technical Note: As RAM evolves from DDR3 to DDR5, the standard operating voltage drops from 1.5V to 1.1V. This reduction is critical for thermal management and battery life in laptops, even as the memory modules operate at significantly higher frequencies.

Task 2: Practical RAM Removal and Installation Guide

Procedure Followed for RAM Maintenance:

Step 1 (Preparation): Completely shut down the PC. Unplug the power cable from the wall and the power supply. Press the power button once to discharge any residual static electricity.

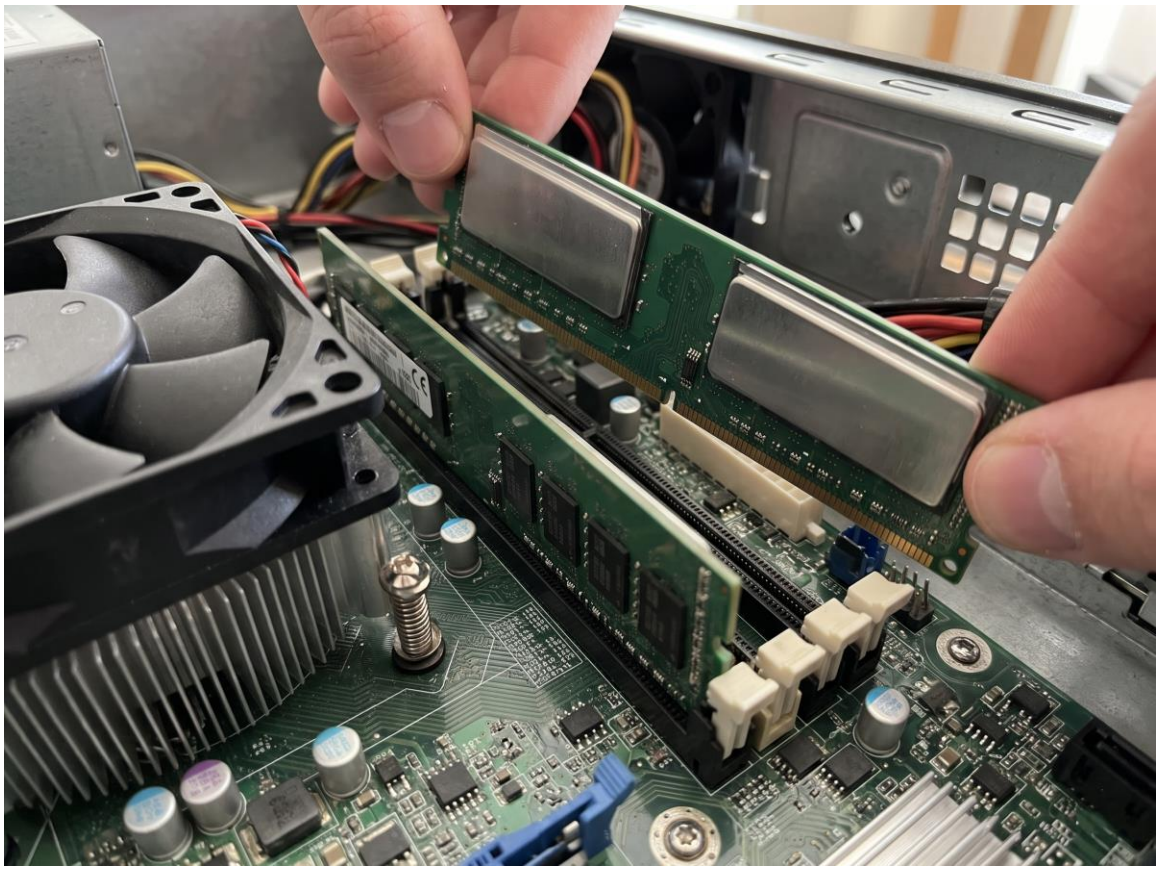
Step 2 (Locating): Open the side panel of the chassis and locate the DIMM slots adjacent to the CPU socket on the motherboard.

Step 3 (Removal): Gently push down the retention clips (tabs) on either side (or just one side, depending on the motherboard) of the RAM module. The stick will pop up slightly. Carefully pull it straight out.

Step 4 (Installation): Align the notch on the bottom of the RAM module with the key in the DIMM slot. Press straight down firmly on both ends until the retention clips snap back into place with a clicking sound.

Step 5 (Verification): Reconnect power and boot the system. Open Task Manager -> Performance -> Memory to verify the full capacity and speed are recognized by Windows.

Evidence of Completion:



Task 3: Key Differences in Storage Types (HDD, SSD, NVMe)

The storage landscape has shifted dramatically from mechanical drives to solid-state flash memory. Below is a detailed breakdown of the three primary storage tiers:

HDD (Hard Disk Drive)

HDDs use mechanical spinning magnetic platters and an actuator arm to read/write data. They max out at around 150-200 MB/s.

• Key Benefit: Extremely low cost per gigabyte, ideal for massive archival storage.

• App Beneficiary: VLC Media Player. Storing a large library of movies or family videos requires terabytes of space. HDDs are perfect for this because playing media does not require high read speeds.

SATA SSD (Solid State Drive)

SATA SSDs use NAND flash memory with absolutely no moving parts, making them durable and silent. They communicate via the SATA III interface, bottlenecking at around 550 MB/s.

• Key Benefit: Huge leap in responsiveness over HDDs; revitalizes older systems.

• App Beneficiary: Google Chrome & Microsoft Office. SSDs drastically reduce application launch times and improve system boot times, making everyday web browsing and document editing feel instantaneous.

NVMe M.2 (Non-Volatile Memory Express)

NVMe drives are a newer form factor (M.2) that plug directly into the motherboard's PCIe lanes, bypassing the slower SATA controller entirely. Speeds can exceed 7,000 MB/s on PCIe Gen 4.

• Key Benefit: Unprecedented read/write speeds for heavy workloads.

• App Beneficiary: Adobe Premiere Pro & Heavy Games (like Cyberpunk 2077 or PUBG). NVMe drives allow 4K video timelines to scrub smoothly without stuttering and enable games to load massive open-world textures in seconds.

Task 4: Practical Storage Drive Inspection

Procedure Followed for Storage Identification:

Step 1 (Safety First): Disconnected all power sources and ensured anti-static precautions were taken.

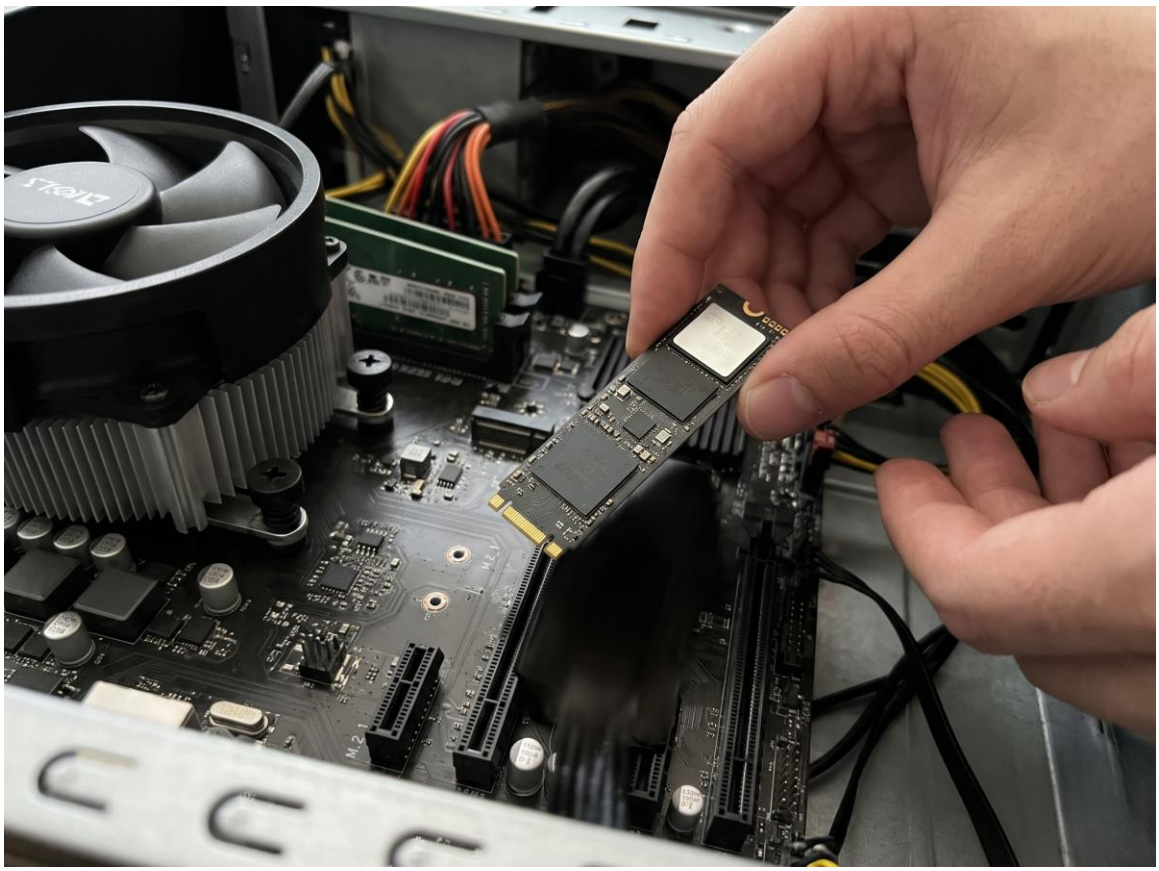
Step 2 (Identification): Inspected the motherboard and drive bays. Located either a 2.5-inch drive bay (for SATA SSD/HDD) connected via SATA data and power cables, OR an M.2 slot flat against the motherboard (for NVMe).

Step 3 (Removal): If SATA, disconnected the two cables. If M.2 NVMe, carefully unscrewed the tiny retention screw and pulled the drive out at a 30-degree angle.

Step 4 (Reinstallation): Reconnected the drive securely to its respective interface.

Step 5 (Verification): Booted into the system BIOS/UEFI to confirm the drive name and capacity were detected in the storage configuration menu.

Evidence of Completion / Connector Type:



https://www.youtube.com/watch?v=fhPYpgLJKtQ